

MÖBIUS INVERSION THROUGH PARTIALLY ORDERED SETS

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1. MÖBIUS FUNCTION

Definition 1.1. A *partially ordered set* (or *poset*) is a pair $(P, \preceq)$ where P is a set and $\preceq$ is a binary relation on P satisfying the following properties for all $x, y, z \in P$:

- (1) **Reflexivity:** $x \preceq x$.
- (2) **Antisymmetry:** if $x \preceq y$ and $y \preceq x$, then $x = y$.
- (3) **Transitivity:** if $x \preceq y$ and $y \preceq z$, then $x \preceq z$.

Lemma 1.2. Suppose that $|P| = n$. Then we can label the elements of $P = \{p_1, p_2, \dots, p_n\}$ so that

$$p_i \preceq p_j \implies i \leq j.^1$$

Proof. We prove this by induction on n . The base case $n = 1$ is trivial. Choose a maximal element of P and label it p_n . Let $P' = P \setminus \{p_n\}$. By the induction hypothesis, we can label $P' = \{p_1, p_2, \dots, p_{n-1}\}$ so that

$$p_i \preceq p_j \implies i \leq j.$$

Since p_n is maximal, this labeling together with p_n gives the required labeling of all of P . □

Let $(P, \preceq)$ be a finite partially ordered set. We define the function

$$\zeta : P^2 \rightarrow \{0, 1\}$$

by

$$\zeta(x, y) = \begin{cases} 1 & \text{if } x \preceq y, \\ 0 & \text{otherwise.} \end{cases}$$

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¹This type of ordering of a poset is called a *topological ordering*.

Definition 1.3. Let P be a finite poset. Consider the matrix

$$A_\zeta = [\zeta(x, y)]_{x, y \in P}.$$

By Lemma 1.2, there is a topological ordering of P , the matrix A_ζ can be an upper triangular matrix with all diagonal entries equal to 1. Hence A_ζ is invertible. Its inverse is denoted

$$A_\mu = [\mu(x, y)]_{x, y \in P}.$$

The function $\mu : P^2 \rightarrow \mathbb{Z}$ is called the *Möbius function* of the poset P .

Since $A_\mu A_\zeta = I$, we obtain

$$\sum_{z \in P} \mu(x, z) \zeta(z, y) = \sum_{x \preceq z \preceq y} \mu(x, z) = \delta(x, y).$$

Theorem 1.4. Consider the two following cases:

(a) Let P be the set of subsets of a finite set X , ordered by inclusion $\subseteq$. Then

$$\mu(A, B) = \begin{cases} (-1)^{|B \setminus A|} & \text{if } A \subseteq B, \\ 0 & \text{otherwise.} \end{cases}$$

(b) Let $P = [n]$ with the order defined by $a \preceq b$ if a divides b . Then

$$\mu(a, b) = \begin{cases} (-1)^r & \text{if } b/a \text{ is the product of } r \text{ distinct primes,} \\ 0 & \text{otherwise.} \end{cases}$$

Proof. We only need to verify the defining relation

$$\sum_{x \preceq z \preceq y} \mu(x, z) = \begin{cases} 1 & x = y, \\ 0 & x \neq y. \end{cases}$$

(a) If $A \subseteq B$, then

$$\sum_{A \subseteq C \subseteq B} (-1)^{|C \setminus A|} = \sum_{k=0}^{|B \setminus A|} \binom{|B \setminus A|}{k} (-1)^k = (1-1)^{|B \setminus A|}.$$

This equals 0 unless $A = B$, in which case it equals 1.

(b) Suppose

$$b/a = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r},$$

where $p_1, p_2, \dots, p_r$ are distinct primes and $k_1, k_2, \dots, k_r \geq 1$. Then any c satisfying $a \mid c \mid b$ corresponds to choosing a subset of the primes $\{p_1, \dots, p_r\}$. Hence

$$\sum_{a \mid c \mid b} \mu(c, b) = \sum_{S \subseteq [r]} (-1)^{|S|} = \begin{cases} 1 & r = 0, \\ 0 & r \geq 1. \end{cases}$$

This verifies the required identity. □

2. MÖBIUS INVERSION

Theorem 2.1. *Let $(P, \preceq)$ be a finite poset and let $f, g, h : P \rightarrow \mathbb{R}$ be functions such that*

$$(2.1) \quad g(x) = \sum_{a \preceq x} f(a), \quad h(x) = \sum_{b \succeq x} f(b).$$

Then the function f can be recovered from g or h via

$$(2.2) \quad f(x) = \sum_{a \preceq x} \mu(a, x)g(a), \quad f(x) = \sum_{b \succeq x} \mu(x, b)h(b).$$

Proof. Since P is finite, we may label its elements as $P = \{p_1, \dots, p_n\}$ in a way compatible with the partial order. Any function $f : P \rightarrow \mathbb{R}$ can then be identified with a column vector

$$\mathbf{f} = (f(p_1), f(p_2), \dots, f(p_n))^T.$$

Similarly we associate vectors $\mathbf{g}$ and $\mathbf{h}$ to the functions g and h .

Let $A_\zeta = [\zeta(x, y)]$ be the matrix corresponding to the zeta function of the poset. The relation

$$g(x) = \sum_{a \preceq x} f(a)$$

means that each coordinate of $\mathbf{g}$ is obtained by summing the entries of $\mathbf{f}$ over elements below it in the order. In matrix form this is

$$\mathbf{g} = A_\zeta^T \mathbf{f}.$$

Likewise, the relation

$$h(x) = \sum_{b \succeq x} f(b)$$

corresponds to the matrix equation

$$\mathbf{h} = A_\zeta \mathbf{f}.$$

Since A_ζ is invertible and its inverse is A_μ , we obtain

$$\mathbf{f} = A_\zeta^{-T} \mathbf{g} = A_\mu^T \mathbf{g} \quad \text{and} \quad \mathbf{f} = A_\zeta^{-1} \mathbf{h} = A_\mu \mathbf{h}.$$

Writing these matrix identities co-ordinatewise yields the stated inversion formulas. $\square$

3. APPLICATIONS OF MÖBIUS INVERSION

3.1. Inclusion–Exclusion Principle. Möbius inversion gives a particularly elegant proof of the classical principle of inclusion–exclusion. Let $\{A_i\}_{i \in I}$ be a family of subsets of a finite set X . For $J \subseteq I$, define

$$f(J) = \left| \left(\bigcap_{i \in J} A_i \right) \cap \left(\bigcap_{i \notin J} (X \setminus A_i) \right) \right|,$$

that is, the number of elements that lie in exactly the sets indexed by J . Let

$$h(J) = \left| \bigcap_{i \in J} A_i \right|.$$

Then

$$h(J) = \sum_{K \supseteq J} f(K).$$

Applying Möbius inversion on the poset $(\mathcal{P}(I), \subseteq)$, we obtain

$$f(J) = \sum_{K \supseteq J} \mu(J, K) h(K) = \sum_{K \supseteq J} (-1)^{|K|-|J|} h(K).$$

Putting $J = \emptyset$ gives

$$\left| \bigcap_{i \in I} (X \setminus A_i) \right| = \sum_{K \subseteq I} (-1)^{|K|} \left| \bigcap_{j \in K} A_j \right|.$$

3.2. Totient Function. Möbius inversion gives another proof of the classical closed form for Euler's totient function. We briefly note that the divisibility relation on the natural numbers forms a partially ordered set. In this *divisibility poset*, we write $a \preceq b$ if a divides b . For a function $f : \mathbb{N} \rightarrow \mathbb{R}$, define

$$g(n) = \sum_{d|n} f(d).$$

Möbius inversion on this poset gives

$$f(n) = \sum_{d|n} \mu(d, n) g(d) = \sum_{\substack{d|n \\ n/d \text{ square free}}} (-1)^{p(n/d)} g(d),$$

where $p(m)$ denotes the number of distinct prime factors of m . For a natural number n , let $\varphi(n)$ denote the number of integers $m \leq n$ such that $\gcd(m, n) = 1$.

Lemma 3.1.

$$n = \sum_{d|n} \varphi(d) = \sum_{d|n} \varphi(n/d).$$

Proof. If $(m, n) = d$, then we can write $m = m_1 d$ and $n = n_1 d$ where $\gcd(m_1, n_1) = 1$. Hence the number of such m is the number of choices of m_1 , which is $\varphi(n_1) = \varphi(n/d)$. $\square$

Applying Möbius inversion to this identity with $g(n) = n$ and $f(n) = \varphi(n)$ gives

$$\varphi(n) = \sum_{d|n} (-1)^{p(n/d)} d = \sum_{d|n} (-1)^{p(d)} \frac{n}{d}.$$

Factoring out n we obtain

$$\varphi(n) = n \sum_{d|n} \frac{(-1)^{p(d)}}{d}.$$

If $n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$ where $p_1, p_2, \dots, p_r$ are distinct primes and $k_1, k_2, \dots, k_r \geq 1$, the multiplicative structure of the sum yields the familiar product form

$$\varphi(n) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right).$$

3.3. Colored Necklace Problem. Möbius inversion also appears naturally in many counting problems. A classical example is the problem of counting colored necklaces.

Suppose we have an infinite supply of beads of λ colours and wish to form a necklace with n beads. Any necklace can be represented by a sequence $(a_1, \dots, a_n)$, where we identify sequences that differ by a cyclic permutation, since they represent the same necklace.

We say that a necklace of length n is *primitive* if it is not obtained by repeating a shorter necklace. More precisely, if $d \mid n$, a necklace is said to have period d if it is obtained by repeating n/d times a primitive necklace of length d .

First note that the number of sequences of length n is λ^n . Every such sequence arises from a primitive necklace of some period d dividing n . If $M(d)$ denotes the number of primitive necklaces of length d , then each such necklace gives rise to d distinct sequences depending on the choice of the starting position. Hence

$$\lambda^n = \sum_{d \mid n} d M(d).$$

Applying Möbius inversion to this identity gives

$$M(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) \lambda^{n/d}.$$

The total number of necklaces of length n is therefore

$$N_n = \sum_{d \mid n} M(d).$$

Substituting the expression for $M(d)$, we obtain

$$N_n = \sum_{d \mid n} \sum_{\ell \mid d} \frac{1}{d} \mu(\ell, d) \lambda^\ell.$$

Writing $d = k\ell$ and rearranging the sums gives

$$N_n = \sum_{\ell \mid n} \frac{\lambda^\ell}{\ell} \sum_{k \mid n/\ell} \frac{\mu(1, k)}{k}.$$

Using the identity

$$\sum_{k \mid m} \frac{\mu(1, k)}{k} = \frac{\varphi(m)}{m},$$

we finally obtain the well-known formula

$$N_n = \frac{1}{n} \sum_{d \mid n} \varphi(n/d) \lambda^d.$$

Thus the number of necklaces of length n formed from λ colours is

$$N_n = \frac{1}{n} \sum_{d \mid n} \varphi(n/d) \lambda^d.$$

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